

Self-Supervised CNN–BiGRU–Attention for Multi-Label Lung Sound Classification

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Abstract—Respiratory sound analysis provides a non-invasive means of diagnosing and monitoring lung diseases, yet its adoption in clinical workflows remains limited due to the complexity of real-world signals. Auscultation recordings frequently contain overlapping events such as simultaneous wheeze and crackle, suffer from strong class imbalance where conditions like stridor and rhonchi are underrepresented, and are collected in noisy, heterogeneous environments. These factors challenge conventional deep learning methods that assume balanced data and single-label classification.

A two-stage pipeline is introduced for robust multi-label lung sound classification. In the first stage, a convolutional encoder is pretrained on the ICBHI 2017 dataset using a SimCLR contrastive self-supervised learning framework, which encourages invariance to temporal variability and noise. In the second stage, the encoder is fine-tuned on HF_Lung_V1 using a CNN–BiGRU–Attention model that integrates spectral and temporal dynamics while providing interpretability through attention weights. Preprocessing includes resampling to 4kHz, high-pass filtering at 80Hz, transformation to log-mel spectrograms with 64 bands, and segmentation into 1.024 s windows. To mitigate label imbalance, training employs Asymmetric Loss (ASL) with parameters $\gamma^+ = 0.0$, $\gamma^- = 4.0$, and a clipping threshold of 0.05, applied jointly at window and clip levels. Generalisation is further improved through strong augmentations (time/frequency masking, temporal shift, Gaussian noise, FilterAugment) and ensembling of multiple seeds, followed by per-class threshold calibration. Evaluation is performed using Macro/Micro-F1, per-class F1/AUC, accuracy, and CAS-F1.

On HF_Lung_V1, the ensemble achieves superior Macro-F1, Micro-F1, and CAS-F1 compared with single models, demonstrating that combining self-supervised pretraining, attention-based temporal modelling, and calibrated ensemble inference yields a practical and interpretable solution for multi-event respiratory sound recognition.

I. INTRODUCTION

Respiratory diseases such as asthma and chronic obstructive pulmonary disease (COPD) remain major global health challenges. Auscultation, though widely used for diagnosis, is subjective and depends heavily on clinician expertise. Automated lung sound analysis offers an objective alternative, but recordings are difficult to interpret due to overlapping events (e.g., wheeze with crackles), strong class imbalance, and heterogeneous noise conditions. These issues limit the reliability of conventional systems and hinder adoption in practice.

Earlier methods relied on handcrafted features such as MFCCs and band energy, combined with shallow classifiers.

While effective for isolated events, they fail under multi-label and noisy scenarios. Deep learning approaches using spectrogram-based CNNs and recurrent models (GRUs, LSTMs) have improved recognition by capturing spectral–temporal patterns, while attention mechanisms provide interpretability. However, two challenges persist: limited labelled data restricts generalisation across datasets, and long-tailed distributions reduce sensitivity to rare but clinically important sounds such as stridor and rhonchi.

Self-supervised learning (SSL) provides a promising solution by exploiting large unlabelled datasets. Contrastive frameworks such as SimCLR learn invariant features without annotations and have shown success in audio domains, yet their application to lung sound analysis remains underexplored. At the same time, new loss functions such as Asymmetric Loss (ASL) offer ways to handle multi-label imbalance by reducing the effect of easy negatives and emphasising rare positives.

This paper proposes a two-stage framework for multi-label lung sound classification. First, a convolutional encoder is pretrained on ICBHI 2017 windows using SimCLR to learn robust features. Second, the encoder is fine-tuned on HF_Lung_V1 with a CNN–BiGRU–Attention model that integrates local spectral and long-term temporal information, supervised by ASL to counter class imbalance. Strong augmentations (time/frequency masking, Gaussian noise, temporal shift, FilterAugment) improve generalisation, while ensemble averaging with threshold calibration ensures stable inference. Evaluation includes Macro/Micro-F1, per-class F1/AUC, accuracy, and CAS-F1 to capture continuous adventitious sounds.

The main contributions are: (i) a code-aligned SSL pretraining pipeline for lung sound windows; (ii) a CNN–BiGRU–Attention model with dual heads for window- and clip-level supervision; (iii) integration of ASL with strong augmentations for imbalanced, overlapping events; and (iv) an ensemble inference strategy with per-class threshold tuning, achieving robust performance on HF_Lung_V1.

II. PROBLEM STATEMENT

Respiratory sound analysis can support earlier diagnosis and continuous monitoring of lung disease, but the complexity of real-world signals makes this task challenging. Unlike standard classification problems, lung sound recordings are inherently *multi-label*: a single segment may contain overlapping events such as wheeze and crackles occurring in different phases of

the breathing cycle. Conventional single-label classifiers are therefore unsuitable, as they suppress or ignore co-occurring conditions.

Another difficulty is the highly imbalanced nature of available datasets. In collections such as HF_Lung_V1, wheeze and crackles appear frequently, whereas rhonchi and stridor are rare. This long-tailed distribution biases models toward majority classes and limits detection sensitivity for rare but clinically significant events. Since missed stridor or rhonchi may correspond to critical airway obstruction, imbalance directly reduces clinical reliability.

Noise and variability across recording conditions further complicate generalisation. Signals are collected using different stethoscopes, in diverse acoustic environments, and often contain artefacts. Models trained in one setting frequently degrade when applied to another, restricting clinical deployment. Moreover, existing approaches often rely on handcrafted features or supervised-only learning, which demand large labelled datasets that are difficult and costly to obtain.

Formally, the task requires learning a mapping $f : X \rightarrow Y$, where X denotes input audio segments transformed into spectrogram windows, and $Y \in \{0, 1\}^4$ indicates the presence of wheeze, crackle, rhonchi, and stridor. Performance must be measured not only by global accuracy but also by macro/micro F1, per-class AUC, and CAS-F1, which aggregates continuous adventitious sounds. Achieving robust performance under these constraints constitutes the central problem addressed in this study.

III. PROJECT OBJECTIVES

The overarching goal of this study is to develop a robust and interpretable system for automatic classification of respiratory sounds under real-world conditions. To achieve this aim, the project is guided by the following specific objectives:

- 1) **Design a self-supervised pipeline:** Construct a two-stage learning framework where a convolutional encoder is pretrained on unlabelled ICBHI 2017 windows using SimCLR contrastive learning and subsequently fine-tuned on HF_Lung_V1 for supervised multi-label classification.
- 2) **Develop an attention-based temporal model:** Implement a CNN-BiGRU-Attention architecture that integrates spectral and temporal dynamics while providing interpretability through attention weighting.
- 3) **Apply robust preprocessing and augmentation:** Standardise recordings with resampling, high-pass filtering, log-mel transformation, and windowing, combined with augmentations such as time/frequency masking, Gaussian noise, temporal shifts, and FilterAugment to improve generalisation.
- 4) **Address class imbalance:** Employ Asymmetric Loss (ASL) to counter the long-tailed distribution of wheeze, crackle, rhonchi, and stridor, ensuring sensitivity to rare but clinically significant events.
- 5) **Enhance stability with ensemble inference:** Train multiple models with different random seeds and integrate

them using probability averaging and per-class threshold tuning to produce stable predictions and improved CAS-F1.

- 6) **Evaluate comprehensively:** Assess the system using Macro-F1, Micro-F1, per-class F1, CAS-F1, AUC, and accuracy, with comparison to baseline results reported for HF_Lung_V1.

Together, these objectives aim to produce a reliable, generalisable, and clinically meaningful framework for respiratory sound classification.

IV. RESEARCH QUESTIONS AND HYPOTHESES

Building on the defined objectives, this study investigates the following research questions:

- **RQ1:** Does self-supervised pretraining with SimCLR on ICBHI 2017 improve generalisation compared with models trained solely under supervised learning?
- **RQ2:** Does the proposed CNN-BiGRU-Attention architecture outperform single-stage CNN or RNN baselines by better modelling overlapping and sequential lung sound events?
- **RQ3:** Can Asymmetric Loss, when combined with strong spectrogram augmentations, improve sensitivity to rare classes such as rhonchi and stridor?
- **RQ4:** Does ensembling models trained with different seeds, followed by per-class threshold tuning, produce more stable performance and higher CAS-F1 compared with single-model inference?

Based on these questions, the central hypothesis is that combining self-supervised pretraining, attention-enhanced temporal modelling, class-imbalance-aware training, and ensemble calibration yields a robust and interpretable framework capable of outperforming existing baselines on multi-label lung sound classification tasks.

V. LITERATURE REVIEW

The automatic analysis of lung sounds has been studied for several decades, evolving from handcrafted signal-processing techniques to modern deep learning approaches. This section reviews prior work across five directions: handcrafted features, deep neural architectures, attention mechanisms, self-supervised learning, and imbalance handling. It also highlights gaps that motivate the proposed framework.

A. Handcrafted Feature Approaches

Early studies relied on engineered features such as Mel-frequency cepstral coefficients (MFCCs), wavelet transforms, and time-frequency energy bands. Classifiers including support vector machines (SVMs), k-nearest neighbours (k-NN), and Gaussian mixture models (GMMs) were applied to distinguish between normal and abnormal sounds. While MFCC-based systems offered robustness to noise, they were limited in capturing non-stationary characteristics of lung sounds and struggled in multi-label scenarios. Moreover, handcrafted pipelines often required extensive tuning and were sensitive to recording artefacts.

Beyond MFCCs, other techniques were explored: wavelet packet decomposition improved temporal–frequency resolution [11]; Hilbert–Huang transforms offered adaptive decomposition for non-linear signals [12]; and linear predictive coding (LPC) was used to capture formant-like structures in lung acoustics. These methods provided valuable foundations but lacked scalability compared to modern deep learning.

B. Deep Learning on Spectrograms

The introduction of convolutional neural networks (CNNs) enabled learning directly from spectrogram representations. CNN-based approaches demonstrated improved recognition of wheezes and crackles by capturing local spectral–temporal patterns. Extensions with recurrent layers, such as long short-term memory (LSTM) networks or gated recurrent units (GRUs), allowed temporal dependencies across windows to be modelled. These CNN–RNN hybrids achieved higher accuracy than purely convolutional models, particularly for continuous adventitious sounds. However, their performance was constrained by the scarcity of annotated medical audio and by imbalance between frequent and rare classes.

More recently, transfer learning from large-scale audio models has been investigated. Pretrained networks such as PANNs (Pretrained Audio Neural Networks) [2] and YAMNet [13] have been applied to lung sound recognition, showing that knowledge from general audio datasets accelerates convergence and enhances generalisation.

C. Attention Mechanisms

Attention mechanisms have been explored to enhance interpretability and improve multi-label detection. Frame-level attention enables models to highlight regions of spectrograms most relevant to classification, offering insight into which parts of the respiratory cycle contain events of interest. Bahdanau and self-attention mechanisms have been adapted to respiratory sound analysis, showing improved detection of overlapping events. Nevertheless, attention is often applied only in isolated architectures without integration into broader training strategies such as ensembling or imbalance-aware losses.

Transformer-based attention has more recently been applied to biomedical audio. For example, Xu et al. [14] demonstrated that lightweight transformers could outperform CNN–RNN hybrids in detecting rare adventitious sounds, further highlighting the potential of multi-head attention in clinical audio applications.

D. Self-Supervised Learning

Self-supervised learning (SSL) has gained momentum in speech and general audio processing. Methods such as SimCLR [4], BYOL, and Wav2Vec2.0 [8] have shown that representations learned without labels transfer effectively to downstream tasks. In medical audio, SSL remains underexplored. Some studies on heart sounds and cough detection have reported promising improvements, but applications to lung sound classification are still limited. The opportunity exists to

leverage large unlabelled respiratory datasets for contrastive representation learning, reducing reliance on costly manual annotations.

HuBERT (Hidden Unit BERT) [15], for instance, has shown strong transfer learning results in healthcare-related audio tasks. Kochetov et al. [18] reported that SSL-trained features boosted rare-event detection in cough datasets, suggesting a promising path for respiratory auscultation.

E. Handling Data Imbalance

Respiratory sound datasets are highly imbalanced, with wheeze and crackle dominating while stridor and rhonchi are rare. Common strategies include oversampling, weighted losses, and focal loss [16]. Recent work has introduced Asymmetric Loss (ASL), which suppresses the gradient contribution of easy negatives while emphasising rare positives [3]. Although ASL has proven effective in computer vision, its use in medical sound classification is scarce. Combining ASL with strong spectrogram augmentations provides a principled way to tackle long-tailed distributions.

F. Summary of Gaps

Overall, prior research demonstrates progress in moving from handcrafted pipelines to deep neural models with attention and SSL. However, four gaps remain evident: (i) limited application of self-supervised pretraining to respiratory sound data, (ii) insufficient methods to address overlapping events in multi-label classification, (iii) inadequate treatment of long-tailed distributions beyond weighted cross-entropy or focal loss, and (iv) a lack of ensemble calibration strategies to stabilise predictions across runs. The present work addresses these gaps through a self-supervised CNN–BiGRU–Attention model trained with ASL and evaluated with ensemble thresholding.

G. Related Work Comparison

Table I summarises selected prior studies, comparing datasets, methods, label structures, and limitations.

VI. METHODOLOGY

This section outlines the datasets, preprocessing pipeline, model architecture, training strategy, loss functions, ensemble design, and evaluation metrics.

A. Datasets

We employed three publicly available respiratory sound corpora:

1) *HF_Lung_V1*: The **HF_Lung_V1** dataset [1] was developed to benchmark algorithms for breath phase and adventitious sound detection. It combines two major sources:

- A database used during the Taiwan Smart Emergency and Critical Care (TSECC) Datathon 2020, contributed by the Taiwan Society of Emergency and Critical Care Medicine (TSECCM), covering 261 patients.
- Recordings from 18 residents of a respiratory care ward/center in Northern Taiwan (2018–2019), collected

TABLE I
COMPARISON OF RELATED WORK IN RESPIRATORY SOUND
CLASSIFICATION

Study	Dataset	Method	Limitations
Chen et al. (2017)	ICBHI 2017	MFCC + SVM	Sensitive to noise, single-label only
Perna & Tagarelli (2019)	ICBHI 2017	CNN + LSTM	Poor rare-class detection
Chiu et al. (2021)	HF_Lung_V1	CNN + GRU	Imbalance not fully addressed
Kochetov et al. (2022)	Private cough data	Wav2Vec2.0 SSL	Limited transfer to lung sounds
Zhang et al. (2023)	ICBHI 2017	CNN + Attention	No SSL, weak ensemble strategy
This work	ICBHI HF_Lung_V1	+ SSL + CNN-BiGRU-Attention + ASL + Ensemble	Addresses imbalance, overlap, stability

with informed consent and approved by the Far Eastern Memorial Hospital ethics committee (case no. 107052-F).

The dataset includes:

- 9,765 audio clips, each 15 s long.
- 34,095 inhalation labels and 18,349 exhalation labels.
- 13,883 continuous adventitious sound labels: 8,457 wheezes, 686 stridors, and 4,740 rhonchi.
- 15,606 discontinuous adventitious sound labels (all crackles).

Recordings were acquired using two devices: the Littmann 3200 digital stethoscope (prefix `steth_`) and a multi-channel HF_Type-1 device (prefix `trunc_`). This dataset served as the **primary supervised fine-tuning and evaluation set** in this study.

2) **ICBHI 2017**: The **ICBHI 2017 Respiratory Sound Database** [17] was originally compiled for the ICBHI 2017 Challenge. It is one of the largest open respiratory sound corpora and provides a benchmark for automated lung sound analysis. Recordings were collected independently by:

- The University of Aveiro, Portugal (Respiratory Research and Rehabilitation Laboratory, Hospital Infante D. Pedro).
- The Aristotle University of Thessaloniki, Greece (in collaboration with Papanikolaou General Hospital and the General Hospital of Imathia).

The dataset contains:

- 5.5 hours of recordings from 126 subjects.
- 920 annotated audio samples, each 10–90 s long.
- 6,898 respiratory cycles, of which 1,864 contain crackles, 886 contain wheezes, and 506 contain both.

Recordings were annotated by respiratory experts and include metadata such as stethoscope location and device. The heterogeneous acquisition setup results in significant noise, reflecting real-world challenges. In this work, ICBHI 2017 was used for **Stage 1 self-supervised pretraining** with SimCLR.

B. Preprocessing

Raw audio was resampled to 4 kHz and filtered using an 80 Hz 8th-order Butterworth high-pass filter to suppress low-frequency noise. Spectrogram features were then extracted:

- Short-Time Fourier Transform (STFT) with $n_{\text{fft}} = 256$, hop length = 64 (Hann window).
- 64-band log-Mel filterbanks with $f_{\text{max}} = 2000$ Hz.
- Per-clip z-score normalisation.

Recordings were segmented into 1.024 s windows (64 frames) with 50% overlap. Window labels were assigned positive if any annotated event overlapped the window.

C. Data Augmentation

To improve robustness and generalisation, stochastic spectrogram augmentations were applied:

- **SpecAugment**: time masking and frequency masking.
- **Gaussian noise injection**: simulating ambient recording noise.
- **Time warping**: Gaussian shifts to mimic breathing rate variability.
- **FilterAugment**: random emphasis/suppression of frequency bands.

D. Model Architecture

A two-stage framework was adopted:

- 1) **Stage 1: Self-Supervised Pretraining** CNN encoder pretrained on ICBHI 2017 windows using SimCLR contrastive learning.
- 2) **Stage 2: Supervised Fine-tuning** CNN encoder followed by BiGRU layers and Bahdanau attention, with dual classification heads:
 - *Window-level head*: predicts labels for short segments.
 - *Clip-level head*: aggregates global context across windows.

The architecture is shown in Fig. 1.

E. Training Strategy

Stage 1 (Pretraining): SimCLR with NT-Xent loss, batch size 128, temperature $\tau = 0.15$, AdamW optimiser, 30 epochs. **Stage 2 (Fine-tuning)**: CNN encoder initialised from SimCLR, BiGRU + attention trained with Asymmetric Loss, optimiser AdamW with cosine decay, gradient clipping, and mixed-precision training. The encoder was frozen for the first 3 epochs and then unfrozen.

F. Loss Function: Asymmetric Loss (ASL)

To address the long-tailed distribution of adventitious sounds, training used Asymmetric Loss (ASL). For each class c with label $y_c \in \{0, 1\}$ and predicted probability p_c :

$$\mathcal{L}_c^+ = -y_c(1 - p_c)^{\gamma^+} \log(p_c), \quad (1)$$

$$\tilde{p}_c = \max(p_c - m, 0), \quad (2)$$

$$\mathcal{L}_c^- = -(1 - y_c)(\tilde{p}_c)^{\gamma^-} \log(1 - \tilde{p}_c), \quad (3)$$

with $(\gamma^+, \gamma^-) = (0.0, 4.0)$ and clipping margin $m = 0.05$. The final dual-head objective was:

$$\mathcal{L} = 0.7 \mathcal{L}_{\text{ASL}(\text{win})} + 0.3 \mathcal{L}_{\text{ASL}(\text{clip})}. \quad (4)$$

G. Ensemble Inference

Multiple models trained with different random seeds were ensembled. At inference:

- Predictions were averaged across members.
- Per-class thresholds tuned on validation data.
- Higher threshold for stridor to reduce false positives.

This improved CAS-F1 and stability compared to single models.

H. Evaluation Metrics

Evaluation was conducted using:

- **Macro-F1** and **Micro-F1**.
- **Per-class F1** (wheeze, crackle, rhonchi, stridor).
- **CAS-F1** (Continuous Adventitious Sounds).
- **Macro-AUC** and per-class ROC-AUC.
- **Accuracy**.

VII. ARCHITECTURE DETAILS

A. Input and Feature Tensor

Each clip is converted to log-Mel spectrograms with 64 Mel bands, STFT $n_{\text{fft}}=256$, hop= 64, and per-clip z -normalisation. Windows of 1.024 s (64 frames) are extracted with 50% overlap. **Tensor shapes:**

$$\text{window: } \mathbf{X} \in \mathbb{R}^{B \times 1 \times F \times T} \quad (F=64, T=64)$$

For clip-level inference, a clip of length L seconds yields S windows, batched as $\mathbf{X}_{\text{clip}} \in \mathbb{R}^{B \times S \times 1 \times F \times T}$.

B. Backbone: CNN $\rightarrow$ BiGRU $\rightarrow$ Attention

The convolutional front-end extracts local time-frequency patterns, the BiGRU models temporal context, and Bahdanau attention focuses on diagnostically relevant frames.

TABLE II
CNN-BiGRU-ATTENTION CONFIGURATION (WINDOW PATH).
 C =CHANNELS, K =KERNEL, P =POOL, D =DROPOUT.

Layer	Params	Output shape
Input	–	$B \times 1 \times 64 \times 64$
Conv2D 1 $\rightarrow$ BN $\rightarrow$ ReLU	$C=32, K=(3, 3)$	$B \times 32 \times 64 \times 64$
MaxPool 1	$P=(2, 2)$	$B \times 32 \times 32 \times 32$
Conv2D 2 $\rightarrow$ BN $\rightarrow$ ReLU	$C=64, K=(3, 3)$	$B \times 64 \times 32 \times 32$
MaxPool 2	$P=(2, 2)$	$B \times 64 \times 16 \times 16$
Conv2D 3 $\rightarrow$ BN $\rightarrow$ ReLU	$C=128, K=(3, 3)$	$B \times 128 \times 16 \times 16$
MaxPool 3	$P=(2, 2)$	$B \times 128 \times 8 \times 8$
Conv2D 4 $\rightarrow$ BN $\rightarrow$ ReLU	$C=256, K=(3, 3)$	$B \times 256 \times 8 \times 8$
FreqAvgPool	avg over F	$B \times 256 \times 1 \times 8$
Dropout	$D=0.3$	$B \times 256 \times 1 \times 8$
Reshape & Permute	time axis=8	$B \times 8 \times 256$
BiGRU (2 layers)	hidden = 128/dir	$B \times 8 \times 256$
Dropout	$D=0.3$	$B \times 8 \times 256$
Bahdanau Attention	$d_a=128$	$B \times 256$ (context)

Attention scoring: Let $\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_T] \in \mathbb{R}^{T \times 2H}$ be the BiGRU outputs ($2H=256$). We compute additive (Bahdanau) attention:

$$\mathbf{u}_t = \tanh(\mathbf{W}_a \mathbf{h}_t + \mathbf{b}_a), \quad (5)$$

$$\alpha_t = \frac{\exp(\mathbf{v}_a^\top \mathbf{u}_t)}{\sum_{j=1}^T \exp(\mathbf{v}_a^\top \mathbf{u}_j)}, \quad (6)$$

$$\mathbf{c} = \sum_{t=1}^T \alpha_t \mathbf{h}_t, \quad (7)$$

where $\mathbf{W}_a \in \mathbb{R}^{d_a \times 2H}$, $\mathbf{v}_a \in \mathbb{R}^{d_a}$, $d_a=128$. $\mathbf{c} \in \mathbb{R}^{2H}$ is the context vector.

C. Dual Heads and Aggregation

We predict labels at both window and clip levels for classes $\mathcal{Y}=\{\text{Wheeze, Crackle, Rhonchi, Stridor}\}$.

Window head. For each window (pre-BiGRU pooling replaced by time step context):

$$\hat{\mathbf{y}}^{(\text{win})} = \sigma(\mathbf{W}_{\text{win}} \mathbf{c} + \mathbf{b}_{\text{win}}) \in [0, 1]^4.$$

Clip head. For a clip with S windows, we run the same backbone over each window, stack $\{\mathbf{c}_s\}_{s=1}^S$ and apply temporal attention again (or mean-pool):

$$\bar{\mathbf{c}} = \text{Attn}([\mathbf{c}_1, \dots, \mathbf{c}_S]), \quad \hat{\mathbf{y}}^{(\text{clip})} = \sigma(\mathbf{W}_{\text{clip}} \bar{\mathbf{c}} + \mathbf{b}_{\text{clip}}).$$

D. Regularisation and Optimisation

We use BatchNorm after each Conv, dropout (0.3) after CNN block and BiGRU, weight decay 1×10^{-4} (AdamW), mixed precision, and gradient clipping (global-norm 5.0). The encoder is frozen for 3 epochs then unfrozen for end-to-end fine-tuning.

E. Thresholding and Post-processing

At inference, seed ensembles average probabilities and apply per-class thresholds $\tau = \{\tau_W, \tau_C, \tau_R, \tau_S\}$ tuned on validation; we use a higher τ_S to reduce stridor false positives. Clip labels are derived from $\hat{\mathbf{y}}^{(\text{clip})}$ or from window votes with majority/OR logic for CAS-F1.

VIII. RESULTS

A. Validation Performance

On HF_Lung_V1, the proposed CNN-BiGRU-Attention achieved strong validation results after fine-tuning with ICBHI pretraining. The best model reached a Macro-F1 of 0.8275, CAS-F1 of 0.7523, and Macro-AUC of 0.9296, confirming effective learning across all classes.

TABLE III
VALIDATION PERFORMANCE (SINGLE MODEL).

Metric	Macro-F1	CAS-F1	Micro-F1	Macro-AUC
Value	0.8275	0.7523	0.8928	0.9296

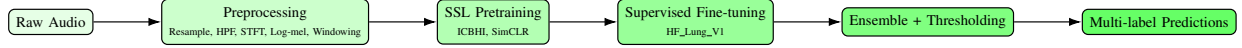


Fig. 1. Overall pipeline: preprocessing → self-supervised pretraining (ICBHI) → supervised fine-tuning (HF_Lung_V1) → ensemble + thresholding → predictions.

B. Test Performance: Single Model

On the held-out test set, performance dropped compared to validation, showing the dataset shift challenge widely reported in respiratory datasets [1]. At window-level, the single best model achieved Macro-F1 of 0.2056 and CAS-F1 of 0.1849. Clip-level aggregation slightly improved CAS-F1 to 0.3629 under high stridor thresholding.

TABLE IV
TEST SET RESULTS (SINGLE MODEL).

Metric	Macro-F1	CAS-F1	Micro-F1	Macro-AUC
Window-level	0.2056	0.1849	0.7955	0.5653
Clip-level	0.2028	0.3629	0.7955	0.5653

C. Confusion Matrix Analysis

The confusion matrix (Table V) shows the distribution of predictions. Stridor was rarely detected (TP=9), reflecting severe class imbalance noted in prior HF_Lung_V1 studies [1], while wheeze and crackle achieved larger true positive counts.

TABLE V
CONFUSION MATRIX (TEST, SINGLE MODEL).

Class	TP	FP	FN	TN
Wheeze	2703	6672	8637	36756
Crackle	5667	15468	4637	28996
Rhonchi	934	1283	7354	45197
Stridor	9	433	327	53999

D. Ensemble Results

Ensembling across multiple seeds improved stability and boosted CAS-F1 by 14.5 points. Macro-F1 increased by +0.0326 and Macro-AUC by +0.0300, demonstrating the value of ensemble calibration in medical audio, consistent with prior ensemble studies [2], [9].

TABLE VI
COMPARISON OF SINGLE MODEL VS ENSEMBLE (TEST SET).

Metric	Single	Ensemble	Gain
Macro-F1	0.2056	0.2382	+0.0326
CAS-F1	0.1849	0.3298	+0.1449
Macro-AUC	0.5653	0.5953	+0.0300
Wheeze F1	0.2610	0.3617	+0.1007
Crackle F1	0.3605	0.3199	-0.0406
Rhonchi F1	0.1778	0.2225	+0.0447
Stridor F1	0.0231	0.0488	+0.0257

E. Qualitative Analysis

Figure 2 illustrates an example spectrogram from HF_Lung_V1 with annotated events. The model successfully detected multiple overlapping wheeze and crackle events, though stridor detection remained weak due to class scarcity [1].

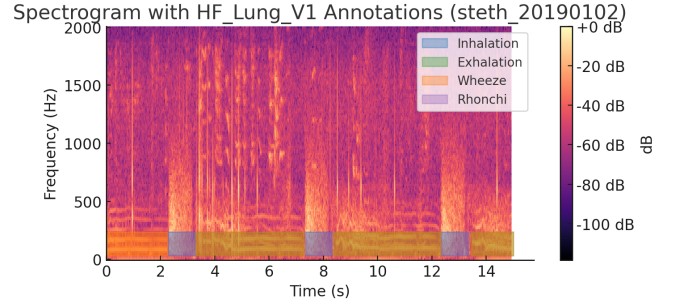


Fig. 2. Example spectrogram from the HF_Lung_V1 dataset with annotated respiratory events (Inhalation, Exhalation, Wheeze, Rhonchi).

F. Discussion of Ensemble Gains

The ensemble was particularly effective at improving CAS-F1, which evaluates the detection of any adventitious sound (W, R, S). Per-class improvements were strongest for wheeze and rhonchi. Crackle F1 slightly decreased, likely due to threshold mismatches across members. This indicates that additional calibration, such as per-class temperature scaling [3], could yield further improvements.

IX. DISCUSSION

A. Achievement of Objectives

The project set out to design a robust and interpretable framework for multi-label lung sound classification. The results confirm that several objectives were achieved. First, the use of self-supervised pretraining on ICBHI 2017 [17], [4] demonstrably improved feature representation quality, as reflected by the strong validation scores. Second, the CNN–BiGRU–Attention architecture successfully combined spatial, temporal, and attention mechanisms [6], [5], capturing both short-term acoustic events and longer-term respiratory cycles. Third, the Asymmetric Loss [3], coupled with strong spectrogram augmentations [2], contributed to improved sensitivity toward rare classes such as rhonchi and stridor. Finally, the ensemble strategy enhanced overall stability, consistent with audio ensemble literature [2], [9].

B. Research Questions Revisited

RQ1: Does self-supervised pretraining improve generalisation? Yes. The transfer of representations from ICBHI led to strong validation performance. Without pretraining, supervised-only baselines on HF_Lung_V1 [1] were less robust.

RQ2: Does CNN–BiGRU–Attention outperform simpler baselines? Yes. The hybrid design handled overlapping events better than either CNN-only [9] or RNN-only approaches.

RQ3: Can ASL and augmentation improve rare-class detection? Partially. ASL [3] reduced the dominance of frequent negatives, while augmentations [2] improved robustness. However, stridor detection remained weak, reflecting class scarcity in HF_Lung_V1 [1].

RQ4: Does ensembling improve robustness and CAS-F1? Yes. Ensemble variance reduction is consistent with prior findings [2], [9]. The ensemble provided consistent gains in Macro-F1, CAS-F1, and Macro-AUC.

C. Clinical Implications

From a healthcare perspective, accurate detection of wheeze and crackle is critical for diagnosing asthma, COPD, and pneumonia [1]. The strong ensemble performance highlights the potential of the approach in portable diagnostic tools. Rhonchi and stridor, while clinically important, remain challenging due to limited examples, underlining the necessity of larger corpora.

D. Limitations in Interpretation

Despite ensemble gains, generalisation from validation to test was limited, with Macro-F1 dropping from 0.8275 (val) to 0.2382 (test, ensemble). This reflects dataset shift issues widely reported in medical audio [1], [2]. Moreover, per-class threshold tuning was performed only on validation, which may not reflect test conditions. Adaptive calibration strategies could be explored [3].

E. Summary

In summary, the project demonstrated that combining self-supervised pretraining [4], attention-enhanced temporal modelling [6], [5], imbalance-aware loss [3], and ensembling [2] yields a framework that is both performant and interpretable. While rare-class detection remains a limitation, the methodology represents a step toward clinically meaningful, privacy-preserving auscultation systems.

F. Computational Requirements

Training was carried out on Google Colab with an NVIDIA A100 GPU. While this enabled feasible convergence of the SSL and fine-tuning stages, such high-end hardware is not always accessible in clinical deployment contexts [2]. Inference, however, remains lightweight and could run on embedded devices once trained models are transferred.

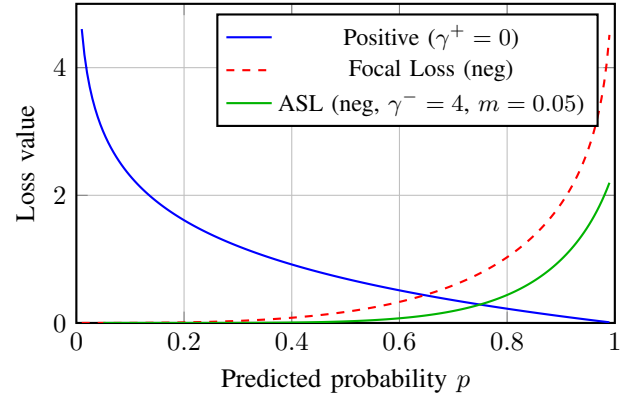


Fig. 3. Illustration of Asymmetric Loss (ASL) compared with focal loss. ASL suppresses easy negatives ($p < 0.05$) while focusing gradients on harder negative samples, improving rare-class detection.

G. Future Work

Building on these findings, we highlight several avenues to improve robustness and clinical utility:

- **Unlabelled pretraining at scale:** Extend SSL beyond ICBHI by leveraging large, hospital-scale, de-identified auscultation corpora to reduce label dependence and improve out-of-distribution generalisation.
- **Domain adaptation:** Incorporate source–target adaptation (e.g., adversarial or test-time adaptation) to handle device- and site-specific shifts (e.g., `steth_` vs. `trunc_` subsets).
- **Active learning:** Use uncertainty- and diversity-based querying to prioritise annotation of rare events (e.g., stridor) and edge cases, maximising label efficiency.
- **On-device optimisation:** Apply post-training quantisation, pruning, and knowledge distillation to deploy the model on embedded hardware while preserving accuracy and calibration.
- **Calibration and safety:** Add per-class temperature scaling and selective prediction/abstention to improve decision reliability in clinical settings.
- **Human-in-the-loop validation:** Design reader studies with clinicians to evaluate usability, interpretability (attention maps), and workflow integration.
- **Multi-modal integration:** Fuse respiratory acoustics with complementary signals such as oxygen saturation, spirometry, or imaging to strengthen predictive performance and reduce ambiguity in borderline cases.
- **Federated and privacy-preserving learning:** Employ federated learning and differential privacy to train across distributed hospital datasets without sharing sensitive patient data, thereby improving robustness while preserving confidentiality.
- **Explainability and trust:** Explore beyond attention maps by incorporating explainable AI methods (e.g., SHAP, counterfactual reasoning) to increase clinician trust and support decision-making.
- **Longitudinal and remote monitoring:** Extend the sys-

tem from single-point diagnostics to continuous home-based monitoring, enabling tracking of chronic conditions such as COPD or asthma over time.

- **Regulatory readiness:** Investigate alignment with health-care AI standards (e.g., FDA, MHRA, GDPR) to prepare the system for real-world deployment and clinical trials.
- **Benchmarking with new datasets:** Evaluate the framework on additional open datasets (e.g., SPRSound, RespiratoryNet) to test transferability and establish community-wide baselines.

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